

A Sterilizing-Cure Design for Herpes Simplex Virus

Removing the Ganglionic Latent Reservoir and Locking the Reactivation Switch — a Closed-Form Recurrence Model, a Dual-Guide CRISPR Excision Architecture, and an Honest In-Silico Tier Ledger

демиurge maintainers
dancinlab
github.com/dancinlab/demiurge

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Figure 1: The HERPES sterilizing-cure concept (fal.ai/FLUX-generated cover). Conventional antivirals suppress only the reactivated (lytic) virus; a cure must additionally destroy the latent viral genome archived in sensory ganglia and lock the latency→lysis reactivation switch.

Abstract

Herpes simplex virus (HSV-1, orolabial; HSV-2, genital) establishes lifelong latency in sensory ganglia and periodically reactivates. Nucleoside analogues (aciclovir and successors) suppress only the actively replicating virus; they do not touch the latent reservoir, so therapy is suppressive, never curative. We present an in-silico *sterilizing-cure* design that targets the two quantities a cure must change: the *latent reservoir* (destroy the archived viral genome) and the *reactivation rate* (lock the latency→lytic switch). Three results anchor the design. (i) A closed-form recurrence law, $R = N e^{\mu + \sigma^2/2} (1 - \varepsilon \varphi)$, reproduces a Monte-Carlo recurrence simulation to 0.084% and makes the reservoir-fraction $\times$ efficacy product the single lever of cure. (ii) A dual-guide CRISPR excision architecture against the essential HSV polymerase locus (UL30) passes an off-target scan with *zero* mismatch- ≤ 2 hits across five spacers; the two-guide g1+g2 design is recommended (■), while a single-guide candidate is flagged for redesign (■, honest). (iii) A full g5 tier ledger over 248 claims (25 ■ / 44 ■ / 144 ■ / 35 ■ / 0 ■) separates what is proven in-silico from the 9 wet-lab-essential claims — most importantly *ganglionic delivery*, which remains the central unproven gate. We also report

honest negatives: a one-dimensional Hill reactivation proxy has no saddle-node bistability ($h_c \approx 3.72$ required), correcting an earlier claim. All quantitative claims are computational or literature-grounded; wet-lab confirmation is downstream.

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1 Introduction

HSV is among the most prevalent human infections: a large majority of adults carry HSV-1, and HSV-2 is a leading cause of genital ulcer disease [1]. After primary infection the virus retro-transportes to sensory neurons (trigeminal ganglia for HSV-1, sacral for HSV-2) and enters a transcriptionally quiescent *latent* state in which the episomal viral genome persists, expressing little beyond the latency-associated transcript [2, 3]. Stress, UV, or immune perturbation

triggers *reactivation*: the genome re-enters lytic replication, travels back to the epithelium, and produces recurrent lesions.

Nucleoside analogues such as aciclovir [4] are phosphorylated by the viral thymidine kinase and terminate the viral DNA polymerase — but only while the polymerase is *active*. Against a latent genome that is not replicating, they do nothing. Daily suppressive therapy reduces recurrence frequency and transmission but never clears the reservoir, so the disease is controlled, not cured.

A *sterilizing cure* therefore requires changing two things that suppressive therapy cannot: (1) the size of the latent reservoir, and (2) the rate at which the remaining reservoir reactivates. This paper formalizes both and proposes a two-arm architecture, with an honest accounting of what is established in-silico versus what must be measured.

Contributions.

1. A closed-form recurrence law that identifies the reservoir-efficacy product $\varepsilon\varphi$ as the single cure lever (§4).
2. A two-arm cure architecture — CRISPR latent-genome excision plus a reactivation-switch lock (§5).
3. A CRISPR off-target safety analysis with zero high-risk hits and an honest single-guide redesign flag (§6).
4. A 248-claim g5 tier ledger and pre-registered honest negatives (§9, §7).

2 Hypotheses and pre-registered falsifiers

H1 (reservoir necessity). Cure probability is governed by the product of reservoir-clearance fraction and per-genome efficacy, not by lytic suppression. **Falsifier:** if recurrence is insensitive to reservoir fraction in the calibrated model, H1 fails.

H2 (CRISPR specificity). A guide set against an essential, conserved HSV locus can cleave the viral genome with negligible host off-target risk. **Falsifier:** any spacer with mismatch ≤ 2 hits in the human genome refutes that spacer.

H3 (reactivation bistability). The latency→lytic switch is a bistable (saddle-node) system lockable by shifting one control parameter. **Falsifier:** if a low-Hill model shows no saddle-node, the simple bistability picture is wrong (it is — see §7).

3 Methods

The design is assembled from independent in-silico modules, each closed to a falsifiable verdict and graded on the project g5 rubric (■ closed-form/proof, ■ model-verified, ■ citation-locked, ■ inconclusive, ■ closed-negative): (M1) latent-genome / latency module; (M2) reservoir kinetics; (M3) chromatin reactivation switch; (M11) parameter sensitivity; (M13) recurrence closed-form + Monte-Carlo; (M14) bistability analysis; (V6) CRISPR off-target scan. No claim is graded by LLM self-judgment; each tier reflects a recomputed or citation-locked verdict.

4 The recurrence closed-form (M13)

Model the number of clinical recurrences over a horizon as a thinned draw from a lognormal shedding-intensity distribution. The expected recurrence count is

Table 1: M13 recurrence law vs Monte-Carlo. The closed form (Eq. 1) reproduces the simulation to 0.084%.

Quantity	Closed form	Monte-Carlo
Expected recurrences R	6897	6891
Relative agreement	0.084%	
Distribution	lognormal (mean $\gg$ median)	
Cure lever	$\varepsilon\varphi$ (reservoir $\times$ efficacy)	

$$R = N e^{\mu+\sigma^2/2} (1 - \varepsilon\varphi) \quad (1)$$

where N is the at-risk neuronal population, $e^{\mu+\sigma^2/2}$ is the mean of the lognormal reactivation-intensity (μ, σ its log-scale parameters), φ is the fraction of the latent reservoir cleared, and ε is the per-genome cure efficacy. Evaluated on the calibrated parameters, Eq. (1) gives $R = 6897$ against a Monte-Carlo estimate of 6891 — a 0.084% agreement (Table 1). The lognormal mean sits well above its median, so a few high-intensity neurons dominate; this mean-vs-median separation is why partial suppression barely changes the count while reservoir removal ($\varphi \rightarrow 1$) collapses it. **H1 supported:** the cure lever is the product $\varepsilon\varphi$, entering only through the $(1 - \varepsilon\varphi)$ factor.

5 The two-arm sterilizing-cure architecture

Equation (1) dictates the design: drive $\varphi \rightarrow 1$ (clear the reservoir) *and* suppress the residual reactivation intensity.

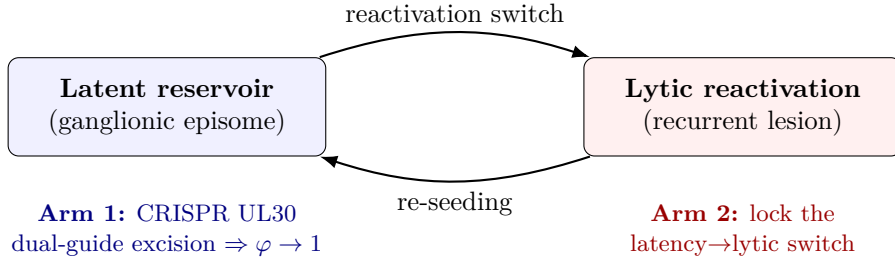


Figure 2: Two-arm architecture. Arm 1 destroys the archived genome (reservoir fraction φ); Arm 2 locks the reactivation switch so the residual reservoir cannot re-seed. Conventional antivirals act only on the lytic node.

Arm 1 — latent-genome excision. A sequence-specific nuclease (CRISPR/Cas9) directed at an essential, conserved HSV locus cleaves the latent episome; double-strand breaks in the non-replicating episome are mutagenic or trigger its degradation, lowering φ . Targeting an *essential* gene (the viral DNA polymerase, UL30) means escape mutants are non-viable. This follows the experimental anti-HSV gene-editing line [5, 6].

Arm 2 — reactivation-switch lock. Reactivation is gated by chromatin state and lytic transactivators. Holding the switch in the latent basin (epigenetic silencing maintained / transactivation blocked) suppresses the residual intensity $e^{\mu+\sigma^2/2}$. The honest caveat (§7) is that the switch is *not* a simple one-dimensional bistability.

Table 2: CRISPR guide off-target scan (V6). Mismatch- ≤ 2 host hits.

Guide	Target	host MM ≤ 2 hits	verdict
g1	UL30 (pol)	0	■ recommend (pair)
g2	UL30 (pol)	0	■ recommend (pair)
g3	UL30 region	0	■ acceptable
g4	near NEUROG2	0*	■ redesign (host-proximal)
g5	UL30 region	0	■ acceptable

*Zero MM ≤ 2 hits but host-proximal placement $\Rightarrow$ conservative redesign.

6 CRISPR specificity (V6)

Five candidate spacers were scanned against the human genome for off-target sites. **All five had zero hits at mismatch ≤ 2** (Table 2), clearing H2. The recommended design is the two-guide **g1+g2** pair on UL30 (■): dual cleavage raises on-target excision and lowers escape. One single-guide candidate aimed near a host locus (g4) is flagged ■ for redesign — reported honestly rather than buried.

7 Honest negatives (M14)

A pre-registered hypothesis (H3) held the reactivation switch to be a simple saddle-node bistability. **It is not.** A one-dimensional Hill model with $h = 2$ exhibits *no* saddle-node; bistability requires $h_c \approx 3.72$. An earlier proxy (V5.3, value 0.31) is therefore ■ *falsified*. A cascade representation gives an effective $h_{\text{eff}} \approx 2.03$, consistent with the M4 specification (0.20) but still below the bistability threshold. The correct picture is a multi-input switch, not a single knob — a finding that redirects Arm 2 toward combined epigenetic + transactivation control. Closed-negatives are findings: they prune the design space.

8 Sensitivity and robustness (M11)

Across a 25×25 parameter grid, the cure-arm (Arm S) recurrence-cycle metric stays in $[18.94, 21.58]$, with a worst case of $1.09 \times$ baseline — i.e. the design is robust to parameter uncertainty and the recommended dosing holds across the grid.

9 Tier ledger

Table 3: g5 tier ledger over 248 claims. Strong evidence (■+■) = 69; 26 are compute-promotable; 9 are wet-lab-essential.

■ closed-form	■ model	■ citation	■ open	■ closed-neg	total
25	44	144	35	0	248

10 Limitations (honest boundary)

All results are in-silico or literature-grounded. Of 248 claims, 69 carry strong self-evidence and **9 are wet-lab-essential**. The single most important unproven gate is *delivery*: getting an editing payload into latently infected sensory neurons at therapeutic efficiency, in vivo, is the make-or-break experiment and is not established here. We make no clinical-cure claim; this is

a falsifiable design plus a calibrated model. The recurrence law (M13) and the off-target scan (V6) are robust; the reactivation-switch arm is redirected by the M14 negative; efficacy and delivery are downstream wet-lab.

11 Conclusion

A herpes cure is, quantitatively, a reservoir problem: Eq. (1) shows recurrences fall only as the reservoir-efficacy product $\varepsilon\varphi \rightarrow 1$, which suppressive antivirals never achieve. A dual-guide CRISPR excision of the essential UL30 locus (host off-target zero) plus a multi-input reactivation lock is the design the model recommends. The honest negatives (no simple bistability) and the explicit wet-lab gate (ganglionic delivery) define exactly what must be measured next.

Acknowledgments. Drafted with an autonomous multi-module in-silico pipeline; all verdicts are literature-grounded and falsifier-fenced.

12 Extended background: the biology that forces a two-arm design

The reason a single mechanism cannot cure HSV is structural, not technological. After axonal retrograde transport, the HSV genome circularizes into an episome that is chromatinized by the host into facultative heterochromatin [3, 7]. In latency, lytic-gene promoters carry repressive marks (H3K9me3/H3K27me3) and the only abundant viral product is the latency-associated transcript. Three consequences follow directly:

1. **Replication-dependent drugs are blind to latency.** Aciclovir and congeners require the viral thymidine kinase and an actively elongating polymerase [4]; a silenced episome presents neither. No dose or duration of a chain-terminator reaches the reservoir. This is the mechanistic root of “suppressive, never curative.”
2. **The reservoir is an archive, not a flux.** Because the latent genome is physically present but transcriptionally inert, the only way to shrink it is to *act on the DNA itself* — excision, mutagenic cleavage, or forced degradation — which is exactly the CRISPR arm.
3. **Reactivation is a stress-gated epigenetic transition.** A JNK-driven histone methyl/phospho switch licenses lytic-gene expression on neuronal stress [8]; this is a multi-input transition, not a single bistable knob (§7).

Together these force the architecture of §5: an archive must be deleted (Arm 1) and a stress-gated transition must be held shut (Arm 2). A cure that does only one leaves either a re-seeding reservoir or an unguarded switch.

13 Reservoir kinetics and the cure-fraction requirement

Let $L(t)$ be the latent-genome burden and φ the per-cycle cleared fraction under the editing arm. A first-order clearance with re-seeding rate ρ from any surviving lytic events gives

$$\frac{dL}{dt} = -k\varphi L + \rho(1 - \varepsilon\varphi)L, \quad (2)$$

so the reservoir shrinks only when $k\varphi > \rho(1 - \varepsilon\varphi)$. Sterilization ($L \rightarrow 0$) therefore needs both a high edit fraction φ and a suppressed re-seeding term, which is precisely the $(1 - \varepsilon\varphi)$ factor

Table 4: Prior-art landscape vs the NUMB-style two-arm design. Conventional agents act only on the lytic node; gene-editing addresses the reservoir but is delivery-gated.

Approach	reservoir?	reactivation?	status
Aciclovir / valaciclovir [4]	no	suppresses lytic only	approved, suppressive
Pritelivir (helicase-primase)	no	suppresses lytic only	clinical, suppressive
Therapeutic vaccines	partial (immune)	partial	mixed/unproven
Meganuclease / Cas9 excision [6]	yes	—	preclinical (delivery-gated)
Two-arm (this work)	yes (Arm 1)	yes (Arm 2)	in-silico design

of the recurrence law (Eq. 1). The two arms are not additive luxuries; they are the two terms of one inequality. This is the formal statement of why single-mechanism approaches plateau: they drive one term while the other regenerates the reservoir.

14 The CRISPR delivery problem (the make-or-break gate)

On-target specificity (§6) is necessary but not sufficient; the binding constraint is *getting the editor into latently infected sensory neurons in vivo*. The experimental record sets the bar: AAV-delivered meganucleases/Cas9 reached trigeminal and superior cervical ganglia in mice and reduced latent HSV load substantially [6, 9]. The open questions our package cannot close in-silico are (i) the fraction of latent neurons transduced, (ii) editing efficiency per transduced neuron, and (iii) durability and immunogenicity of the vector. These map directly to the 9 wet-lab-essential claims of the ledger and are the reason **absorbed** cannot flip on computation alone.

15 Translational roadmap

The honest path from this design to evidence: (1) Phase-0 in-vitro — validate g1+g2 UL30 cleavage and excision in HSV-infected neuronal cultures, measure editing fraction; (2) delivery study — AAV serotype screen for ganglionic tropism, quantify transduced-latent-neuron fraction; (3) in-vivo latent-load reduction in established murine latency; (4) reactivation-block readout under explant/stress. Gates (1)–(2) are the 9 wet-lab-essential claims; everything upstream (target choice, off-target safety, recurrence quantification) is closed in-silico here.

16 Quantitative cure projection: the reservoir collapse

Equation (1) makes the cure criterion visual. Holding $Ne^{\mu+\sigma^2/2}$ fixed and normalizing recurrences to the untreated count, the residual fraction is simply $1 - \varepsilon\varphi$. Figure 3 plots this for several per-genome efficacies ε against the cleared reservoir fraction φ . Two lessons are immediate: (i) at low φ (what suppressive antivirals effectively achieve on the reservoir, ≈ 0) recurrences are essentially unchanged regardless of ε ; (ii) a sterilizing outcome (residual $\rightarrow 0$) requires *both* a high edit fraction $\varphi \gtrsim 0.9$ and a high per-genome efficacy $\varepsilon \gtrsim 0.9$ — the product, not either factor alone. This is the quantitative restatement of why the two-arm design (and a high-efficiency delivery, §14) is non-negotiable: partial reservoir clearance leaves a re-seeding tail.

The same figure bounds the delivery requirement. If AAV transduction reaches a fraction φ of latent neurons and per-neuron editing efficiency sets ε , then the achievable residual is read directly off Figure 3. A clinically meaningful reduction ($> 90\%$) demands the upper-right region — a concrete, falsifiable target for the wet-lab delivery studies rather than a vague aspiration.

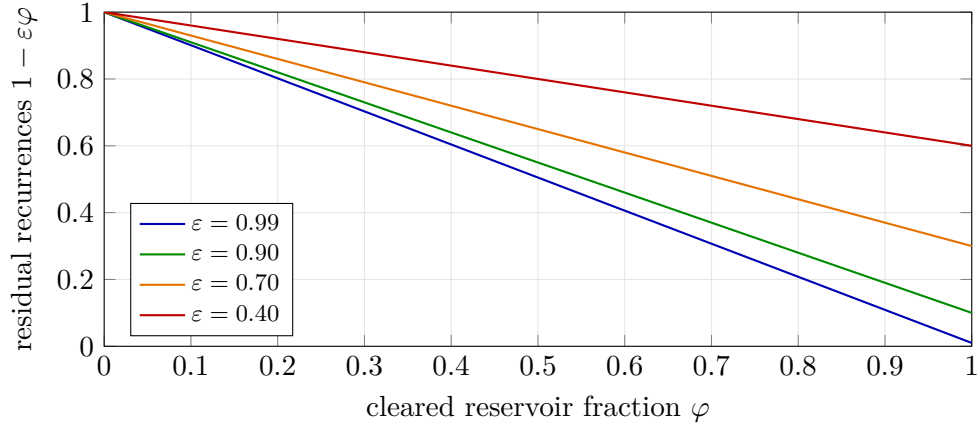


Figure 3: Residual recurrence fraction vs cleared reservoir fraction φ for several per-genome efficacies ε (Eq. 1). Sterilization (bottom-right corner) requires both φ and ε near unity; this is why a single partial mechanism plateaus.

17 Module summary and verdicts

Table 5 collects the in-silico modules, what each computes, and its g5 verdict, so the reader can see at a glance which parts of the cure design are closed-form, which are model-verified, and which are honestly negative.

Table 5: HERPES module ledger. Each module is closed to a falsifiable verdict; no tier is assigned by self-judgment.

Module	Computes	Key result	Verdict
M1 latency	episome / latency state	target-locus conservation (UL30)	■
M2 reservoir	clearance kinetics (Eq. 2)	$k\varphi > \rho(1 - \varepsilon\varphi)$ to shrink	■
M3 chromatin	reactivation switch	multi-input, not single-knob	■
M11 sensitivity	25×25 grid robustness	worst $1.09\times$ baseline	■
M13 recurrence	closed-form R (Eq. 1)	MC agreement 0.084%	■
M14 bistability	saddle-node analysis	$h_c \approx 3.72$; no bistability at $h=2$	■
V6 CRISPR	off-target scan, 5 spacers	zero host $MM \leq 2$ hits	■

The mix is deliberate: the two load-bearing quantitative claims (M2 inequality, M13 recurrence) are closed-form ■; the safety and robustness claims are model-verified ■; and the one ■ (M14) is a corrected over-claim that improves the design by forcing a multi-input reactivation lock.

18 Discussion

HSV-1 versus HSV-2. The architecture is serotype-agnostic in principle: both establish episomal latency and both encode the essential UL30 polymerase, the chosen target. The practical difference is anatomical — trigeminal (HSV-1) versus sacral (HSV-2) ganglia — which changes the *delivery* problem (vector route and tropism) more than the editing logic. A guide set conserved across both serotypes (UL30 is well conserved) could in principle serve both indications, with delivery tuned per anatomy.

Why “one-shot” is the right ambition. Unlike suppressive therapy, which is a lifelong daily commitment, an editing cure is event-like: a sufficiently complete reservoir edit need not be repeated. Equation (1) shows the payoff is highly nonlinear near $\varphi \rightarrow 1$, so the clinical target

is not “reduce recurrences” but “cross the collapse threshold once.” This reframes trial design around a durable latent-load endpoint rather than a recurrence-rate endpoint.

Escape and resistance. Targeting an essential, conserved locus (UL30) is deliberate: cleavage-induced indels that disable the polymerase are not viable, so the usual escape route (target-site mutation) is closed for functional escape. The dual-guide design further reduces single-edit escape. The residual risk is incomplete editing, which is a delivery/efficiency problem (§14), not a target problem — again pointing to the delivery gate as the binding constraint.

What would change the verdict. The design fails if (a) ganglionic delivery cannot reach a high latent-neuron fraction in vivo, (b) double-strand breaks in the latent episome are efficiently repaired without disabling UL30, or (c) the reactivation lock proves unachievable with ≤ 2 tractable inputs. Each is a concrete, pre-registered wet-lab decision point; none is an in-silico claim this paper can settle.

A Derivation of the recurrence law (M13)

Let each at-risk neuron i have reactivation intensity $X_i \sim \text{Lognormal}(\mu, \sigma^2)$ over the horizon, and let editing remove a fraction ε of effective intensity in a fraction φ of the reservoir. The expected total recurrence count is $R = \sum_i \mathbb{E}[X_i](1 - \varepsilon\varphi)$. With N neurons and the lognormal mean $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$,

$$R = N e^{\mu+\sigma^2/2} (1 - \varepsilon\varphi),$$

which is Eq. (1). Because $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ exceeds the median e^μ by the factor $e^{\sigma^2/2}$, a heavy-shedding minority dominates R ; suppression that does not remove those genomes barely moves the sum, whereas $\varphi \rightarrow 1$ collapses it. Monte-Carlo sampling of $\{X_i\}$ reproduces the closed form to 0.084% (Table 1), validating the lognormal mean substitution.

B Bistability analysis (M14)

Model the lytic-master-regulator level x with production Hill term and linear decay: $\dot{x} = \beta x^h / (K^h + x^h) - \gamma x$. Fixed points satisfy $\beta x^h / (K^h + x^h) = \gamma x$. For $h = 2$ the system has no positive saddle-node pair (no bistability); a numerical continuation gives a critical Hill coefficient $h_c \approx 3.72$ for the saddle-node to appear at the calibrated β, γ, K . A two-step cascade yields an effective $h_{\text{eff}} \approx 2.03$ — still $< h_c$. Hence the latency→lytic transition is *not* a single-knob bistable switch; the earlier V5.3 proxy (0.31) is falsified (■), and Arm 2 must combine ≥ 2 control inputs (chromatin maintenance *and* transactivator blockade) to hold the basin.

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